HERIOT-WATT UNIVERSITY

F79MB Statistical Models B

Assessment Project 2 – 2025: Mark scheme

1. (a) Summary statistics for the price of the vehicle are:

```
Min.   1st Qu.   Median    Mean   3rd Qu.    Max.
281     6925     13992     17882   25925     69999
```

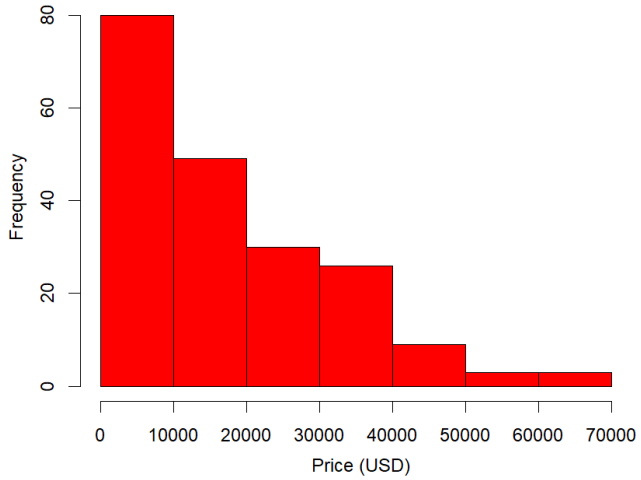


Figure 1: Histogram of prices (in USD) for 200 observations.

Figure 1 has a standard deviation of $14,335. The histogram of vehicle prices shows a unimodal, highly right-skewed distribution, which is consistent with the summary statistics, where the mean of $17,882 is significantly greater than the median of $13,992.

Summary statistics for the age of the vehicle are:

```
Min.   1st Qu.  Median   Mean    3rd Qu.   Max.
2.00    5.00    9.00     10.18   13.00     85.00
```

Figure 2(i) shows that vehicle age (as of 2022) follows a highly right-skewed distribution. According to the summary statistics, 75% of the vehicles are less than 13 years old. An outlier is observed at 85 years, likely representing an antique vehicle. The standard deviation of vehicle age is 7.78 years.

Figure 2(ii) shows that Ford is the most common brand in the dataset, accounting for 59 vehicles. Toyota and Chevrolet follow closely, each with approximately 46 vehicles, while Nissan and Honda have lower counts at 27 and 21, respectively.

For fuel type, Figure 2(iii) indicates that most vehicles run on gas (86%), with a smaller proportion classified as 'other' (9.5%) and an even smaller share using diesel (4.5%).

Regarding vehicle type, Figure 2(iv) highlights that sedans make up the largest category (40%), while pickups (31.5%) and SUVs (28.5%) have comparable distributions.

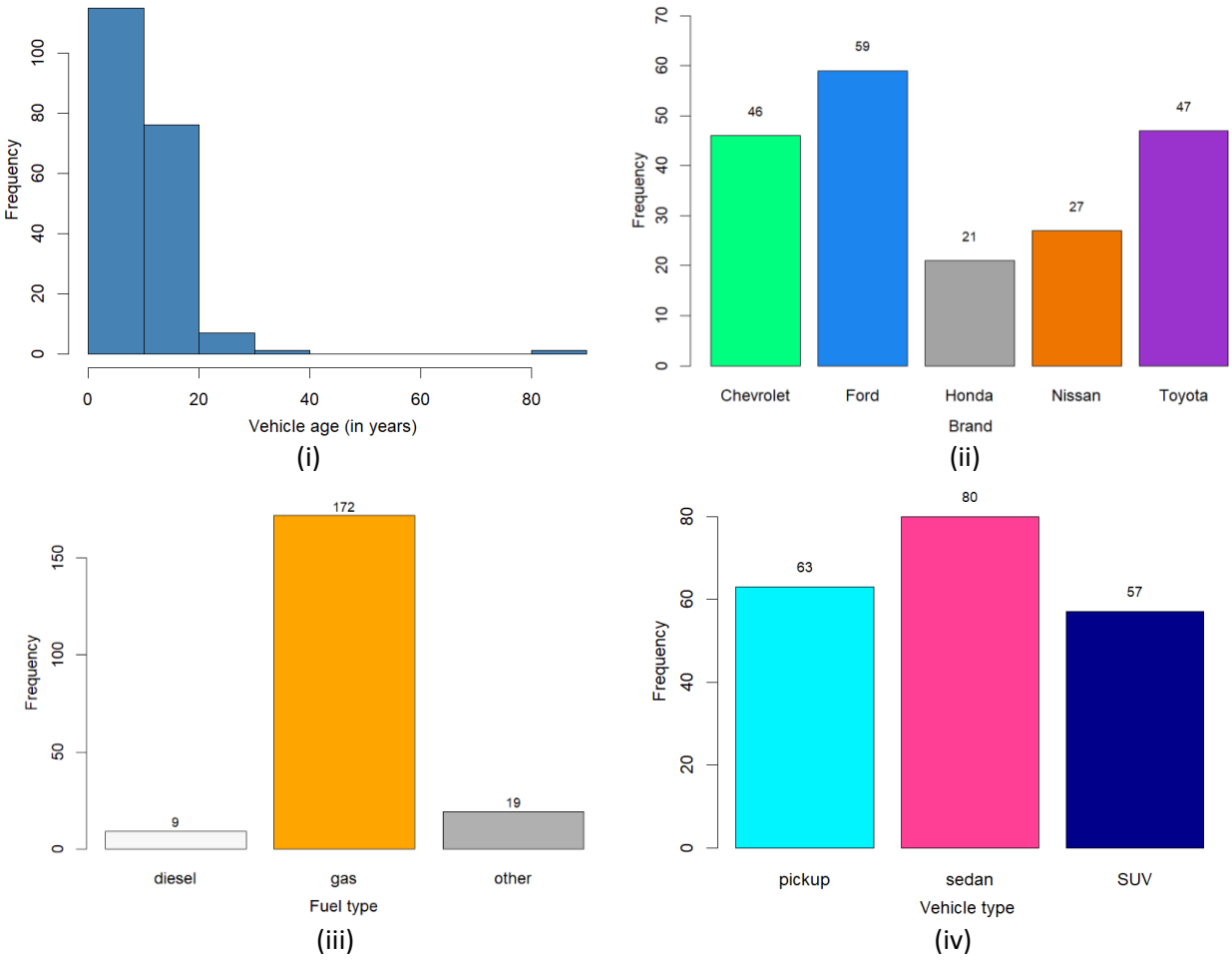


Figure 2: Distribution of data for (i) vehicle age (top left), (ii) brand (top right), (iii) fuel type (bottom left), and (iv) vehicle type (bottom right).

1. (b) The boxplots in Figure 3(i) show that vehicle prices across different brands are right-skewed. The median price varies by brand, with Ford and Toyota having higher medians and a wider price range, while Nissan tends to have a smaller range and comparatively lower prices.

The scatter plot of price against vehicle age in Figure 3(ii) shows a clear downward trend, indicating that prices generally decrease as vehicle age increases, with a significantly steeper decline for newer cars, suggesting a non-linear relationship. Most vehicle ages are under 20 years, but there is an outlier at a vehicle age of 85.

Figure 3(iii) boxplots show that the vehicle types for 'other' and 'diesel' appear normally distributed, with diesel vehicles tending to have the highest prices among the car types, while gas vehicles are skewed to the right, with four outliers, and exhibit a higher price deviation (broad distribution), with many lower-priced cars.

As seen in Figure 3(iv), pickup trucks are slightly right-skewed and generally have higher median prices compared to sedans and SUVs. Both sedans and SUVs are right-skewed, but the price range of sedans is the smallest. Notably, there are outliers, particularly for SUVs, which have five outliers compared to one for sedans. SUVs also exhibit a larger price spread compared to pickups and sedans.

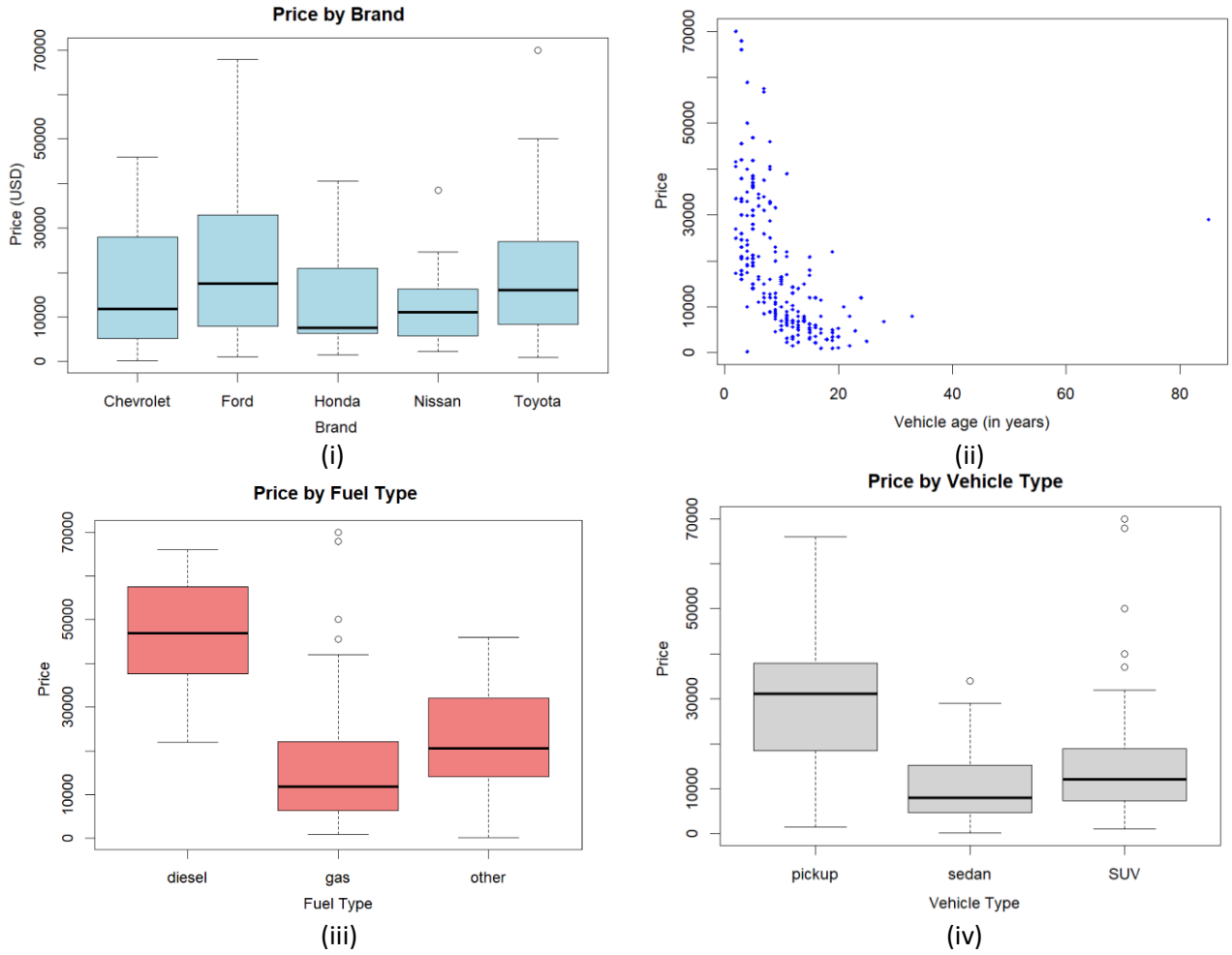


Figure 3: Plots of relationships between vehicle price and (i) brand, (ii) vehicle age, (iii) fuel type, and (iv) vehicle type.

1. (c) The Pearson correlation coefficient between price and age is estimated from the data of size $n = 200$, yielding $r = -0.41398$, which suggests a moderate negative linear relationship. Assuming bivariate normality, a 99% C.I for ρ based on Fisher's transformation is given by:

$$\left(\tanh\left(\tanh^{-1}(r) - Z_{0.005}\sqrt{\frac{1}{n-3}} \right), \tanh\left(\tanh^{-1}(r) + Z_{0.005}\sqrt{\frac{1}{n-3}} \right) \right).$$

Coding this in *R* gives the CI **$(-0.5539, -0.2514)$**. Since the interval does not include zero, there is evidence of significant negative correlation between price and age at the 99% confidence level.

However, Fisher's method assumes bivariate normality, which appears to be violated based on the scatter plot in Figure 3(ii), as vehicle prices exhibit a non-linear decreasing trend with vehicle age. If the data follow a bivariate normal distribution, the scatter plot should be approximately elliptical. Additionally, the linear relationship itself is questionable, as observed in the scatter plot. Given this potential non-linearity, the computed CI may not be entirely reliable.

A more suitable approach could involve non-parametric methods, such as Spearman's rank correlation or bootstrapped confidence intervals, which do not rely on the assumption of linearity.

1. (d) Fitting the proposed model (**Model 1**), which includes price as the response variable and the interaction between age and type, but excludes interactions with fuel type and brand, we obtain the below summary output results:

```
lm(formula = price ~ age*type + fueltype + brand, data=vehicles)

               Estimate    Std.Error  t value    Pr(>|t|)
(Intercept)    53788.8     3631.6      14.811    < 2e-16 ***
age            -1396.3      174.1      -8.020    1.10e-13 ***
typesedan     -27731.9     2588.5     -10.714    < 2e-16 ***
typeSUV        -4800.6     3222.3      -1.490    0.13795
fueltypegas   -17105.9     3331.7      -5.134    7.04e-07 ***
fueltypeother -12093.4     3809.2      -3.175    0.00175 **
brandFord       1266.1     1819.2       0.696    0.48732
brandHonda      2730.5     2487.5       1.098    0.27375
brandNissan    -1548.4     2209.3      -0.701    0.48428
brandToyota     3408.2     1857.0       1.835    0.06803 .
age:typesedan   1338.5      201.9       6.631    3.44e-10 ***
age:typeSUV     -215.0      276.6      -0.777    0.43786

Residual standard error: 8897 on 188 degrees of freedom
Multiple R-squared:  0.6361,  Adjusted R-squared:  0.6148
F-statistic: 29.88 on 11 and 188 DF,  p-value: < 2.2e-16
```

The R^2 value of 0.6361 indicates that Model 1 explains 63.61% of the variability in the price, with the remaining 36.39% of the variation is unexplained by the model. Significant predictors include **age** ($p < 1.1\times10^{-13}$), **typesedan** ($p < 2\times10^{-16}$), **fueltypegas** ($p = 7.04\times10^{-7}$), **fueltypeother** ($p = 0.00175$), and the interaction **age:typesedan** ($p = 3.44\times10^{-10}$). On the other hand, **typeSUV** ($p = 0.13795$), brand variables (***p* > 0.06**), and **age:typeSUV** ($p = 0.43786$) do not statistically significant contribute to the model at the 5% significance level.

To gain a clearer understanding, we consider the analysis of variance (ANOVA) results for Model 1, as shown below:

```
             Df   Sum Sq     Mean Sq     F value  Pr(>F)
age           1   7.008e+09   7.008e+09   88.548   < 2e-16 ***
type          2   1.008e+10   5.040e+09   63.680   < 2e-16 ***
fueltype      2   2.811e+09   1.406e+09   17.758   8.62e-08 ***
brand         4   5.631e+08   1.408e+08    1.779   0.135
age:type      2   5.551e+09   2.776e+09   35.070   1.14e-13 ***
Residuals   188   1.488e+10   7.915e+07                     <
```

We observe that age, type, fuel type, and the interaction terms between age and type are all highly significant ($p < 8\times10^{-8}$) , while brand ($p = 0.135$) which is not statistically significant. Next, we fit **Model 2** by removing the brand variable. The results are as follows:

```
lm(formula = price ~ age*type + fueltype, data=vehicles)

               Estimate    Std. Error  t value   Pr(>|t|)
(Intercept)    55340.6     3228.3      17.142    < 2e-16 ***
age            -1403.2      174.7      -8.034    9.36e-14 ***
typesedan     -27652.2     2514.6      -10.997    < 2e-16 ***
typeSUV        -4959.6     3239.3      -1.531    0.12740
fueltypegas   -17420.6     3264.9      -5.336    2.66e-07 ***
fueltypeother -12268.2     3759.9      -3.263    0.00131 **
age:typesedan   1342.5      203.2       6.606    3.79e-10 ***
age:typeSUV     -197.2      277.5      -0.711    0.47816

Residual standard error: 8964 on 192 degrees of freedom
Multiple R-squared:  0.6228,  Adjusted R-squared:  0.609
F-statistic: 45.28 on 7 and 192 DF,  p-value: < 2.2e-16
```

The analysis of variance (ANOVA) results for Model 2:

```
             Df   Sum Sq      Mean Sq     F value  Pr(>F)
age           1   7.008e+09   7.008e+09   87.23    < 2e-16 ***
type          2   1.008e+10   5.040e+09   62.73    < 2e-16 ***
fueltype      2   2.811e+09   1.406e+09   17.49    1.05e-07 ***
age:type      2   5.568e+09   2.784e+09   34.65    1.42e-13 ***
```

```
Residuals    192    1.543e+10    8.034e+07
```

From the summary output of Model 2, all terms are highly significant, except for **typeSUV** (p = 0.1274) and **age:typeSUV** (p = 0.47816) do not statistically significant. The ANOVA table shows that age, type, fuel type, and the interaction terms between age and type are all highly significant ($p < 1\times10^{-7}$) and should be retained in the model.

Upon examining the residuals plots for Model 2 in Figure 4, the the 'Residuals vs Fitted' plot suggests potential non-linearity, possibly a quadratic pattern. The 'Normal Q-Q' plot shows deviations from normality, particularly in the upper tail. The 'Scale-Location' plot shows a fairly random scatter, suggesting no major concerns regarding heteroscedasticity. However, the 'Cook's Distance' plot identifies observation 39 as an influential point, with a Cook's distance value of 11.113, indicating a significant impact on the model.

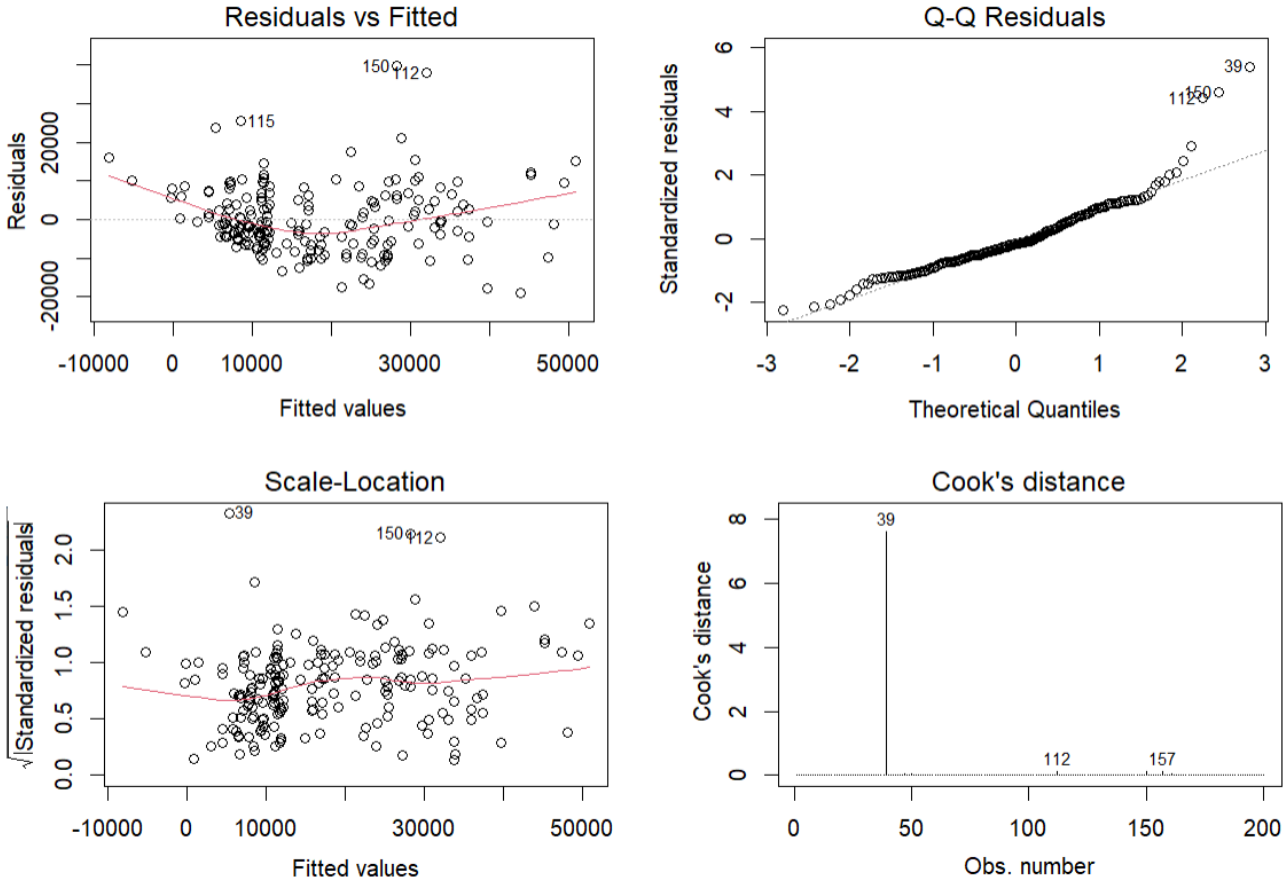


Figure 4: Plots of residuals for the regression of Model 2.

It seems reasonable to remove observation 39 and re-analyze the reduced data, as this influential point corresponds to a vehicle age of 85 years, which does not seem sensible or useful to include in the fitted model to predict price. Therefore, its removal appears justified. After excluding this observation, we fit **Model 3**, the summary output and the corresponding analysis of variance (ANOVA) results are as follows:

```
lm(formula = price ~ age*type + fueltype, data = vehicles_reduced)

                Estimate    Std.Error  t value   Pr(>|t|)
(Intercept)     55401.0     2982.9     18.573    < 2e-16 ***
age             -1411.8      161.4     -8.748    1.13e-15 ***
typesedan      -17952.2     2859.1     -6.279    2.24e-09 ***
typeSUV         -5374.9     2993.9     -1.795    0.0742 .
fueltypegas    -17065.7     3017.3     -5.656    5.59e-08 ***
fueltypeother  -14192.4     3489.8     -4.067    6.96e-05 ***
age:typesedan     368.4      251.5      1.465    0.1446
```

```
age:typeSUV      -188.5     256.4     -0.735    0.4630

Residual standard error: 8282 on 191 degrees of freedom
Multiple R-squared:  0.6786,  Adjusted R-squared:  0.6669
F-statistic: 57.62 on 7 and 191 DF,  p-value: < 2.2e-16
```

The analysis of variance (ANOVA) results for Model 3:

```
             Df    Sum Sq     Mean Sq     F value    Pr(>F)
age           1    1.565e+10  1.565e+10   228.200    < 2e-16 ***
type          2    9.511e+09  4.755e+09    69.327    < 2e-16 ***
fueltype      2    2.215e+09  1.108e+09    16.148    3.31e-07 ***
age:type      2    2.890e+08  1.445e+08     2.107    0.124
Residuals   191    1.310e+10  6.859e+07
```

From the summary output of Model 3, all terms are highly significant except for **typeSUV** (p = 0.0742), **age:typesedan** (p = 0.1446), and **age:typeSUV** (p = 0.463), which are not statistically significant at the 5% level. The ANOVA table indicates that age, type, and fuel type are highly significant ($p < 3.3\times10^{-7}$) and should be retained in the model. However, the interaction terms between age and type is insignificant (p = 0.124), suggesting they may not contribute meaningfully to the model.

Next, we fit **Model 4**, excluding both the interaction terms between age and type as well as observation 39. The summary statistics for Model 4 results are as follows:

```
lm(formula = price ~ age + type + fueltype, data = vehicles_reduced)

              Estimate    Std.Error  t value   Pr(>|t|)
(Intercept)    55022.4    2874.8     19.140    < 2e-16 ***
age            -1357.7     106.5    -12.752    < 2e-16 ***
typesedan     -14241.2    1476.9     -9.643    < 2e-16 ***
typeSUV        -7507.8    1613.7     -4.652    6.08e-06 ***
fueltypegas   -17061.3    3025.9     -5.638    6.03e-08 ***
fueltypeother -14947.6    3477.2     -4.299    2.72e-05 ***

Residual standard error: 8330 on 193 degrees of freedom
Multiple R-squared:  0.6716,  Adjusted R-squared:  0.663
F-statistic: 78.92 on 5 and 193 DF,  p-value: < 2.2e-16
```

All terms for **age**, **type**, and **fuel type** are highly significant ($p < 2.7\times10-5$), suggesting that these variables should be retained in the model. Therefore, there is no need for further ANOVA analysis. The F-test provides strong evidence ($p < 2.2\times10^{-16}$) that Model 4 fits significantly better than the null model. Furthermore, the R^2 value of 0.6716 indicates that Model 4 explains 67.16% of the variability in price, capturing a substantial portion of the price variation.

Now, considering the residuals plots for Model 4 shown in Figure 5. The 'Residuals vs Fitted' plot suggests potential non-linearity, possibly indicating a quadratic pattern. The 'Normal Q-Q' plot shows deviations from normality, with some points in the upper tail lying above the reference line and standardized residuals exceeding 5, the 'Scale-Location' plot indicates no major signs of heteroscedasticity, and the 'Cook's Distance' plot shows no influential points.

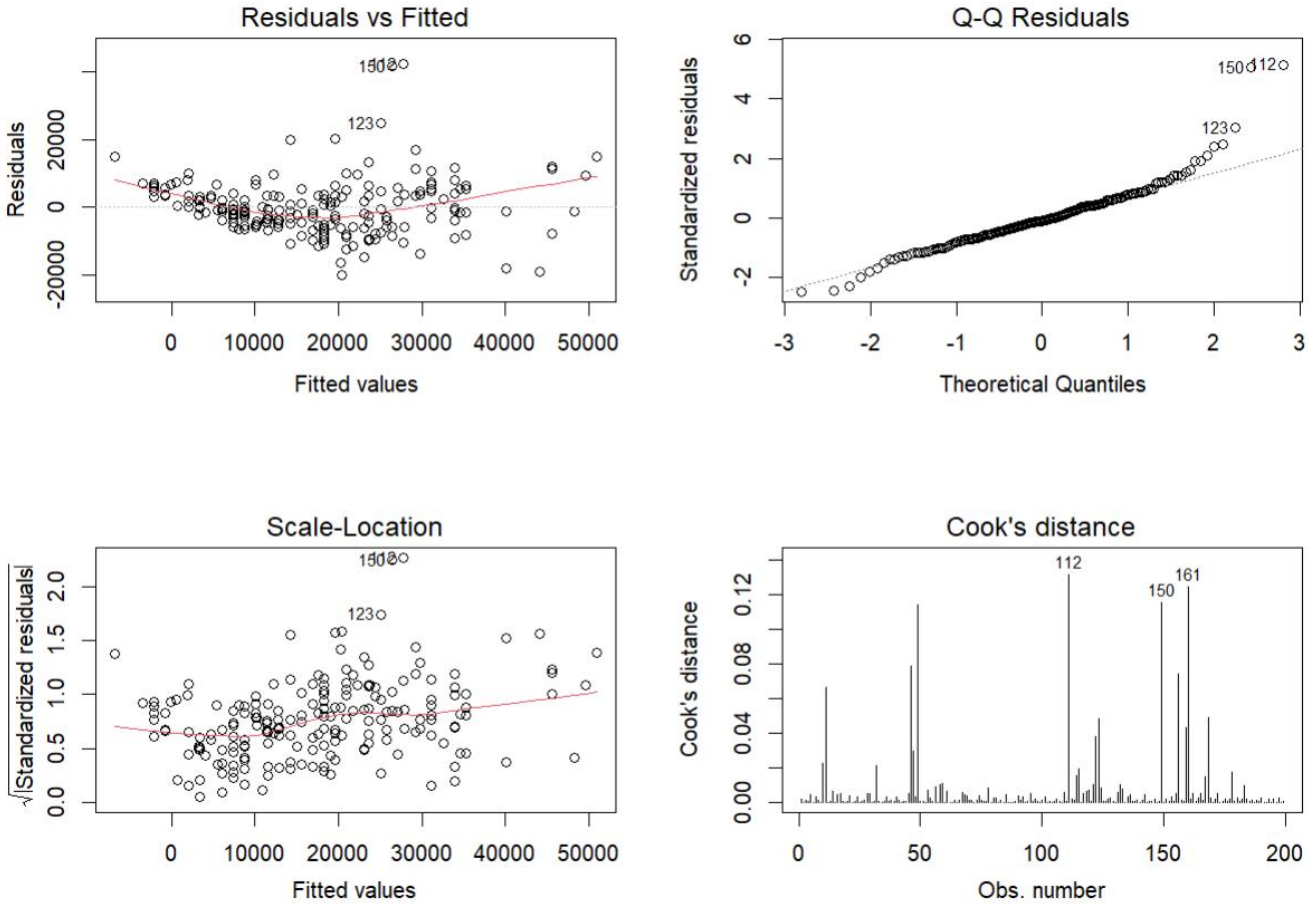


Figure 5: Plots of residuals for the regression of Model 4.

Thus, Model 4 (without the interaction terms between age and type and excluding observation 39) is preferred and its fitted model is

$$\text{Price}_i = 55022.4 - 1357.7(\text{age}_i) - 14241.2 \times I\{\text{type} = \text{sedan}\} - 7507.8 \times I\{\text{type} = \text{SUV}\} \\ -17061.3 \times I\{\text{fueltype} = \text{gas}\} - 14947.6 \times I\{\text{fueltype} = \text{other}\}$$

where $I(\,)$ is the indicator function, with 'pickup' as the base level for 'type' and 'diesel' as the base level for 'fueltype'.

Interpretation: Model 4 shows that each additional year of a vehicle's age reduces its price by $1,357.7, assuming other factors stay the same. Sedans and SUVs are estimated to cost $14,241 and $7,507 less than pickups, respectively, with age and fuel type unchanged. Also, gas-powered vehicles are estimated to be $17,061 cheaper, and vehicles with other fuel types are $14,947 cheaper than diesel-powered vehicles, considering age and type.

Conclusion: The preferred Model 4, which includes the explanatory variables age, type, and fuel type (excluding the interaction term and observation 39), provides valuable insights into vehicle price, with both the *F*-statistic and all coefficient *t*-tests being highly significant. The fitted model suggests that as a vehicle's age increases, its price decreases. Additionally, the type of vehicle, particularly sedans and SUVs, negatively impacts the price compared to pickups, while fuel type (gas and other) also leads to lower prices compared to diesel-powered vehicles. With an R^2 value of 0.6716, the model explains approximately 67% of the variability in price based on these explanatory variables. However, the R^2 value is not very high compared to models with values above 80%, suggesting there is room for improvement. Also, there are concerns with the model diagnostics. The 'Residuals vs Fitted' plot suggests a potential non-linear pattern. The non-linear pattern, as seen in Figure 3 (ii), seems expected, given that the scatter plot shows a potentially exponential decrease in price as age increases. The assumption of normality appears to be imperfectly met. Furthermore, as shown in Figure 6, the plot of

fitted price versus actual price reveals that many predicted values do not closely align with the actual prices. In conclusion, while Model 4 provides useful insights, there is still room for further refinement.

Discussion of further analyses for refining the model:

1. **Adding more variables**: The R^2 value of approximately 0.67 indicates that the model explains about 67% of the variation in vehicle price, leaving room for improvement. Vehicle price could depend on additional factors that are not currently included in the model, such as engine size/power, vehicle mileage, and vehicle condition, among others.
2. **Non-linear modelling**: The residuals plot and scatter plot of price vs. age suggest a potential non-linear relationship between age and price, possibly an exponential or quadratic trend. This indicates that a more complex model may better capture the relationship between these variables. Some approaches to address this includes adding polynomial terms (e.g., age^2) , exploring non-linear regression models, and other techniques.
3. **Transformation of variables**: The issues with normality in the residuals suggest that transformations of certain variables may be helpful. For example, applying a logarithmic transformation to the dependent variable (price) or independent variables (such as age) may stabilize variance and help meet normality assumptions.
4. **Use of Alternative Models**: The current model assumes normality, homoscedasticity and linearity, which may not fully hold. To improve the model's performance, alternative modelling techniques could be considered such as spline regression, generalized least squares (GLS), and machine learning methods could be considered.

By adding more variables, exploring non-linear relationships, applying transformations, and considering alternative models, the model can be improved to better explain vehicle price and increase predictive accuracy.

[**Remark:** Since the sample already includes $n = 200$, collecting additional data may not be the most effective way to improve the model – unless a clear justification is provided that the dataset lacks representation of certain vehicle types, price ranges, or market conditions.]

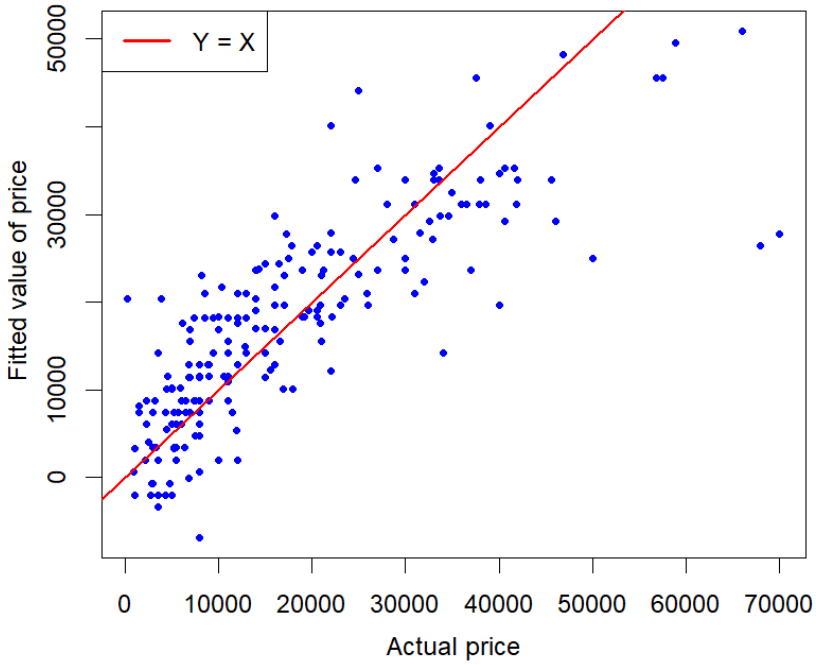


Figure 6: Scatter plot of fitted versus actual vehicle prices for Model 4.

2. (a) From Figure 7, we observe that Business class has the highest proportion of '1 = Satisfied' passengers, followed by Economy class, while Economy Plus class shows the lowest satisfaction, likely due to its small sample size. Both Economy and Economy Plus classes have a significantly higher proportion of '0 = Neutral or Dissatisfied' passengers compared to satisfied ones, suggesting that most passengers in these classes tend to be neutral or dissatisfied.

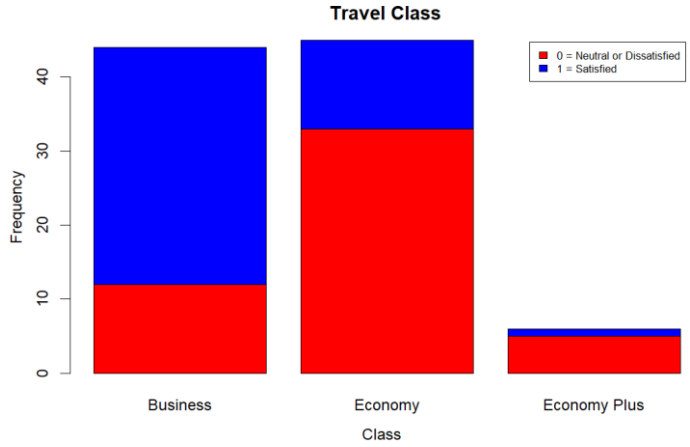


Figure 7: Stacked barplot of travel class by satisfaction status.

From Figure 8, the boxplots show the distribution of travel distances (in km) for both satisfaction categories: '0 = Neutral or Dissatisfied' and '1 = Satisfied'. The distribution for '1= Satisfied' has a noticeably larger range and appears more right-skewed compare to '0 = Neutral or Dissatisfied'. However, both boxplots have no outliers. This suggests that passengers who travelled longer distances were more likely to be satisfied.

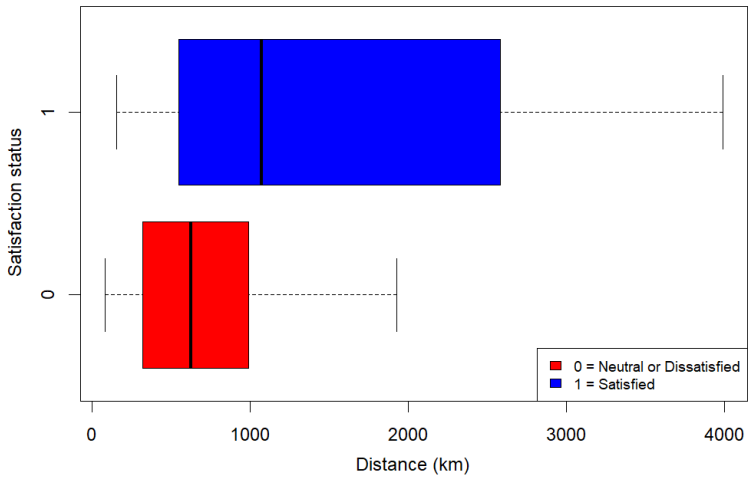


Figure 8: Boxplots of travel distances (Distance) by satisfaction status (Satisfaction) for a sample of 95 airline passengers.

2. (b) Table 1 below shows the contingency table of satisfaction status and gender. We use the χ^2 test of independence to test

H_0: Satisfaction and Gender are independent
H_1: Satisfaction and Gender are not independent

Table 1: Contingency table of satisfaction status and gender for a sample of 95 airline passengers.

Satisfaction	Gender		Row Total
	Female	Male	
0 (Neutral or Dissatisfied)	30	20	50
1 (Satisfied)	28	17	45
Column Total	58	37	95

The expected frequencies under H_0 are

Satisfaction	Gender	
	Female	Male
0	30.526	19.474
1	27.474	17.526

Since the expected frequency in each cell is greater than 5, there is no need to use Fisher's exact test. With Yates' continuity correction, the test statistic χ^2 =0.00012297 on 1 df, the associated *p*-value is $P(\chi_1^2 > 0.00012297)$= 0.9912. That is, there is no evidence of an association between satisfaction status and gender.

Table 2 below shows the contingency table of satisfaction status and travel class. As observed, some cells contain very small numbers, with some having counts as low as 1. Due to limitations of conducting a χ^2 test of independence with these low counts, Fisher's exact test is a suitable alternative.

Table 2: Contingency table of satisfaction status and travel class of 95 airline passengers.

Satisfaction	Class			Row Total
	Business	Economy	Economy Plus	
0 (Neutral or Dissatisfied)	12	33	5	55
1 (Satisfied)	32	12	1	45
Column Total	44	45	6	95

We use the Fisher test of independence to test:

H_0: Satisfaction and Class are independent
H_1: Satisfaction and Class are not independent

From the *R* output, the associated *p*-value for the Fisher's exact test is 9.922×10^{-6}. This suggests strong evidence of an association between satisfaction status and travel class.

2. (c) For a generalised linear model with binomial response, we choose the default link function of logistic link. Fitting a GLM with logistic link and all 4 explanatory variables (Gender, Class, Distance, Delay) without interaction terms (Model GCDL), we obtained the following output:

```
                Estimate Std. Error z value Pr(>|z|)
(Intercept)   -0.3348030  0.6193991  -0.541  0.58883
GenderMale    -0.1331083  0.5058675  -0.263  0.79245
ClassEco      -1.3969749  0.5216531  -2.678  0.00741 **
ClassEco Plus -2.0644959  1.1654388  -1.771  0.07649 .
Distance       0.0008917  0.0003611   2.470  0.01353 *
Delay          0.0111908  0.0093929   1.191  0.23349

    Null deviance: 131.43  on 94  degrees of freedom
Residual deviance: 100.37  on 89  degrees of freedom
AIC: 112.37
```

The individual hypothesis tests indicate that the coefficients for GenderMale and Delay are not significant (*p*=0.79245 and *p*=0.23349). The coefficient ClassEco Plus is significant at the 10% level but not at the 5% level. The other coefficients ClassEco and Distance are both significant at the 5% level (*p*=0.00741 and *p*=0.01353).

Analysis of Deviance for Model GCDL gives the following results.

```
        Df  Deviance  Resid.Df  Resid. Dev  Pr(>Chi)
NULL                     94      131.44
Gender   1   0.0492      93      131.38       0.824443
```

```
Class     2  22.4332      91       108.95       1.345e-05 ***
Distance  1   7.1027      90       101.85       0.007697  ***
Delay     1   1.4754      89       100.37       0.224499
```

It shows that Gender is highly insignificant (p=0.824443); in the presence of Gender, the inclusion of Class is highly significant (p=1.345×10^{-5}); then in the presence of Gender and Class, the inclusion of Distance is also significant (p=0.007697); However, in the presence of Gender, Class, and Distance, the inclusion of Delay is insignificant (p=0.224499). Therefore, it is suggested to exclude Gender for further investigation.

Next, we fit below models separately, first without the Gender explanatory variable, then without the Class explanatory variable, and so on, along with the null model. The following residuals deviance and AIC values were obtained.

Model	**Terms present**	**Deviance**	**df**	**AIC**
Model GCDL	Gender + Class+ Distance +Delay	100.37	89	112.37
Model CDL	Class+ Distance +Delay	100.44	90	110.44
Model GCD	Gender + Class+ Distance	101.85	90	111.85
Model CD	Class+ Distance	101.93	91	109.93
Model C	Class	109.16	92	115.16
Model D	Distance	112.19	93	116.19
Model N	Null	131.43	94	133.43

Analysis of deviance:
Comparing Model GCDL with Model CDL, change in deviance is 100.44 – 100.37 = 0.069405 on 1 df, with the p-value, $p=P(\chi_1^2 > 0.069405)$=0.7922. Removing the Gender term from the model does not significantly reduce goodness-of-fits.

Comparing Model GCDL with Model GCD, change in deviance is 101.85 – 100.37 = 1.4754 on 1 df, with the p-value, $p= P(\chi_1^2 > 1.4754)$ =0.2245. Removing the Delay term from the model does not significantly reduce goodness-of-fits.

Comparing Model GCDL with Model CD, change in deviance is 101.93 – 100.37 = 1.5572 on 2 df, with the p-value, $p=P(\chi_2^2 > 1.5572)$=0.459. Removing both Gender and Delay terms from the model does not significantly reduce goodness-of-fits.

Now, comparing Model CD with Model C, change in deviance is 109.16 – 101.93 = 7.2316 on 1 df, with the p-value, $p=P(\chi_1^2 > 7.2316)$=0.007163. That is, removing the Distance term results in a significant reduction in goodness-of-fits.

Comparing Model CD with Model D, change in deviance is 112.19 – 101.93 = 10.261 on 2 df, with the p-value, $p=P(\chi_2^2 > 10.261)$=0.005914. That is, removing the Class term results in a significant reduce in goodness-of-fits.

From analysis of deviance results, the Class and Distance terms should be retained in the model. Thus, we prefer Model CD.

Comparing Model CD with the null model, change in deviance is 131.44 – 101.93 = 29.503 on 3 df, with the p-value, $p=P(\chi_3^2 > 29.503)$=1.755×10^{-6}. So, strong evidence that the model with Class and Distance (Model CD) fits better than the null model.
Comparing Model CD with the maximal model, the residual deviance is 101.9313 on 91 df, with the associated p-value, $p=P(\chi_{91}^2 > 101.9312)$=0.20358. No evidence against goodness-of-fit of the model.

That is, the Model CD with explanatory variables Class and Distance is not significantly worse than the maximal model.

Looking at the AIC values, Model CD has the lowest AIC of 109.93. The differences in AIC between Model CD and other models are as follows: 2.46 for Model GCDL (112.37 − 109.93), 5.23 for Model C (115.16 − 109.93), and 6.26 for Model D (116.19 − 109.93). Since these differences exceed the rule-of-thumb threshold of 2, this suggests that Model CD provides a substantial improvement over Model GCDL, Model C and Model D.

Overall, Model CD with explanatory variables Class and Distance included is preferred. The fitted model is

$$\ln\left(\frac{\hat{p}}{1-\hat{p}}\right) = -0.1335 - 1.46\, I\{Class = Eco\} - 2.15\, I\{Class = Eco\ Plus\} + 0.00084 Distance \quad (1)$$

where $\hat{p}$ is the estimated probability of a passenger being satisfied, and $I\{\}$ is the indicator function. The base level for Class is Business. A passenger is classified as '1 = Satisfied' if $\hat{p} \geq 0.5$; otherwise, they are classified as '0 = Neutral or Dissatisfied'.

The fitted model (1) shows that, holding Distance variable constant:

- The odds of satisfaction for Business class relative to Economy class is $e^{-1.46} = 0.232$, meaning that the odds for Economy class are 76.8% lower that for Business class.
- The odds of satisfaction for Business class relative to Economy Plus class is $e^{-2.15} = 0.116$, meaning that the odds for Economy Plus class are 88.4% lower that for Business class.

The coefficient for Distance indicates that, holding Class fixed, there is 0.084% increase in the odds of a passenger is satisfied for each additional kilometre travelled, since $e^{0.00084} = 1.00084$.

Figure 9 is the plot of deviance residuals against fitted values for Model CD. The residuals are in two groups, which is expected to be for binomial data with only two possible outcomes (Satisfied, or neutral or dissatisfied). A clear pattern in the residuals plot shows a feature of binary response variable Satisfaction, rather indicating a deficiency with the fit.

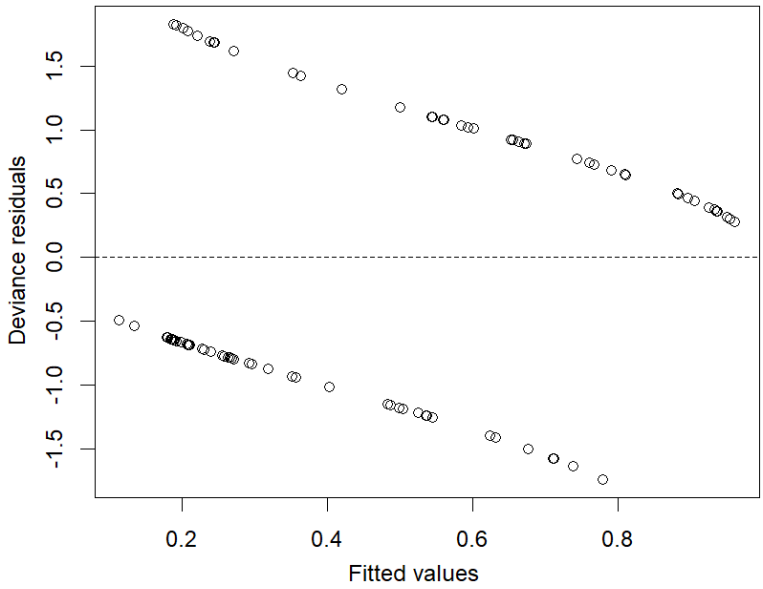


Figure 9: Deviance residuals against fitted values for Model CD.

Conclusion: The analysis of deviance and AIC values suggest that the explanatory variables Class and Distance significantly influence airline passenger satisfaction, while Gender and Delay did not significantly affect the probability of satisfaction. Based on hypothesis tests and model comparisons,

Model CD, which includes Class and Distance as predictors, is preferred over more complex models incorporating Gender and Delay.

The AIC value of 109.93 is the lowest among all models, with differences exceeding the threshold of 2, confirming its superiority. The residual deviance plot for Model CD suggests no major model inadequacy. The preferred fitted Model CD indicates that Business class passengers are more likely to be satisfied compared to Economy and Economy Plus passengers, and longer travel distances slightly increase the likelihood of satisfaction (see Figure 10). These insights can inform airline service improvements and targeted strategies for enhancing passenger experiences.

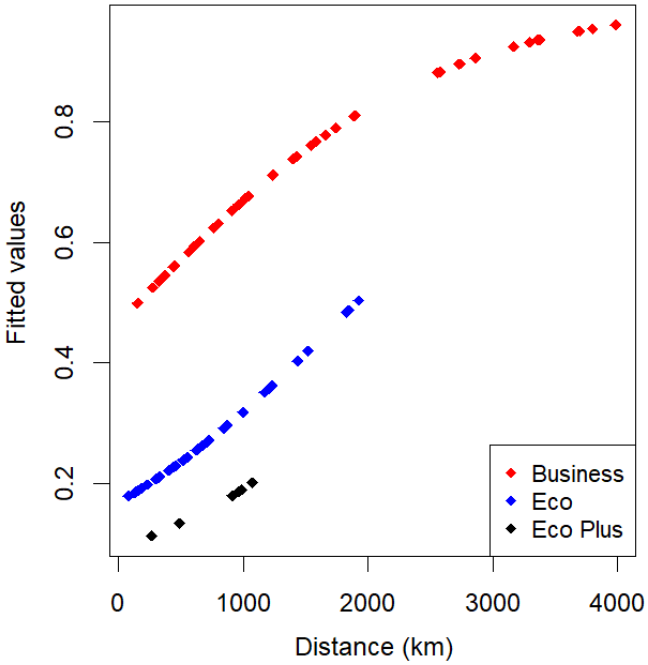


Figure 10: Fitted values plotted against distance by travel class in Model CD.

2. (d) Under the preferred Model CD, the predicted probability of a passenger being satisfied is calculated using equation (1). For example, for a passenger travelling in Economy class over a distance of 160 km (first data), the probability of satisfaction is

$$\hat{p} = \frac{e^{-0.1335-1.46(1)+0.00084(160)}}{1+e^{-0.1335-1.46(1)+0.00084(160)}} = 0.1878.$$

This estimated probability of 0.1878 (which is less than 0.50) indicates that a passenger travelling 160 km in Economy class is classified as '0 = Neutral or Dissatisfied'. However, according to the given data, the actual satisfaction status is '1 = Satisfied', which shows an inaccurate in prediction for the first dataset.

When the predictions for the first 10 dataset are evaluated, as shown Table 3, the accuracy is 60% (6 out of 10 correct predictions).

Table 3: Predicted probabilities and satisfaction status for the first 10 airline passenger entries using Model CD.

Dataset No.	$\hat{p}$	**Predicted Satisfaction**	**Actual Satisfaction**	**Class**	**Distance (in km)**
1	0.18776	0	1	Eco	160
2	0.90569	1	1	Business	2863
3	0.19188	0	0	Eco	192

4	0.93657	1	1	Business	3377
5	0.35217	0	1	Eco	1182
6	0.20780	0	1	Eco	311
7	0.96093	1	1	Business	3987
8	0.88135	1	1	Business	2556
9	0.24356	0	1	Eco	556
10	0.79014	1	1	Business	1744

While a 60% accuracy rate is decent, there is room for improvement. The misclassification for passengers in the first 10 entries, such as the passenger in Economy class travelling 160 km being predicted as 'Neutral or Dissatisfied' (when they were actually 'Satisfied'), indicates that the model may not be fully capturing the complexity of the factors influencing passenger satisfaction.

The model's prediction accuracy could potentially be improved in several ways:

1. **Include more variables**: The current model may be missing key predictors that could improve accuracy. Variables like Inflight Wifi service, food and drink quality, seat comfort, baggage handling, inflight entertainment, and etc could provide additional context that influences satisfaction.
2. **Interaction terms**: Adding interaction terms between variables might help the model account for combinations of factors that influence satisfaction, such as class and distance or seat comfort and inflight entertainment.
3. **Nonlinear relationships**: The current model uses a linear form for the predictor variables, which may not capture the full complexity of relationships. Using nonlinear models or transformations (e.g., polynomial features or splines) could better fit the data.
4. **Model alternative**: Consider experimenting with other classification models, such as decision trees, random forests, or gradient boosting models, which may capture more complex patterns in the data that logistic regression may not fully address.
5. **Model calibration**: The threshold for classifying predictions (e.g., 0.5 for satisfied vs. neutral/dissatisfied) could be adjusted based on performance evaluation, such as using precision-recall curves, or through cross-validation. This could help align the predicted probabilities with actual outcomes more accurately.

By exploring these enhancements, the model's ability to predict passenger satisfaction could improve, leading to more accurate predictions overall.

Rubric Exposition/Presentation

Overall exposition/presentation: 10 marks. Follow the rubric in Appendix B. Note that this rubric is written with the intention that 10 marks will only be awarded to an exceptionally good piece of work; expect most students to get 8 to 5 marks; expect that fewer students will get 0 to 4 marks.